Haokai Ding

Shanghai, China

+86 133 0193 8580 | ditang0125@gmail.com | haokaiding.github.io

Summary

Undergraduate researcher in robotics, focusing on underactuated grasping, idle-stroke and Peaucellier-based linkages, and contact-rich aerial manipulation. Building hardware prototypes and analytical models that bridge mechanism design with field-deployable manipulation.

Education

Tsinghua University | Shenzhen X-Institute — *Shenzhen, China*

TSIEN EXCELLENCE IN ENGINEERING PROGRAM (TEEP) – X SCHOLAR — *Sep. 2023 – Present*

- Joint training program in underactuated robotics and mechanism design.

Shenzhen Technology University – Future Technology School — *Shenzhen, China*

B.ENG. IN ELECTRONIC SCIENCE AND TECHNOLOGY — *Sep. 2022 – Jul. 2026*

- GPA: 85.08/100. Research focus: LIBS spectroscopy and robotics systems.

Research Experience

State Key Lab of Mechanical Systems & Vibration, Shanghai Jiao Tong University — *Shanghai, China*

VISITING STUDENT – UAV PERCHING AND AERIAL MANIPULATION — *Feb. 2026 – Present*

- Advisor: Prof. Wei Dong.
- Working on UAV perching and contact-rich aerial manipulation.

Shenzhen X-Institute, Tsinghua University — *Shenzhen, China*

UNDERGRADUATE RESEARCHER – UNDERACTUATED ROBOTIC GRIPPERS — *2023 – Present*

- Designed and tested underactuated grippers based on semi-Peaucellier linkages.
- Developed idle-stroke and delay-triggering mechanisms; validated prototypes via simulation and physical experiments.

Shenzhen Technology University — *Shenzhen, China*

UNDERGRADUATE RESEARCHER – LIBS SPECTROSCOPY FOR HEAVY-METAL ANALYSIS — *2022 – 2023*

- Built and optimized a LIBS system for rapid heavy-metal adsorption analysis.
- Improved the calibration pipeline and acquisition accuracy.

Publications

† Co-first author * Corresponding author

SELECTED CONFERENCE PAPERS (ROBOTICS)

[P1] A Novel Gripper with Semi-Peaucellier Linkage and Idle-Stroke Mechanism for Linear Pinching and Self-Adaptive Grasping
H. DING, W. ZHANG* — *IEEE/RSJ IROS 2025*

[P2] Semi-Peaucellier Linkage and Differential Mechanism for Linear Pinching and Self-Adaptive Grasping
H. DING, Z. CHEN, T. YANG*, W. ZHANG* — *IEEE CASE 2025*

[P3] BioCrest – A Flexible Adaptive Robotic Hand Based on Idle-Stroke Mechanism
H. DING, X. ZHONG, W. HUANG, T. YANG*, W. ZHANG* — *IEEE ROBIO 2025*

[P4] GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation
H. DING, H. LUAN, T. YANG*, W. ZHANG* — *IEEE ICCMA 2025*

[P5] Peaucellier Gripper: A Novel Underactuated Gripper for Linear Pinching and Self-Adaptive Grasp
H. DING, J. FAN, Z. ZHU, Y. ZHAO, K. CHEN*, W. ZHANG* — *IEEE ICARM 2025*

[P6] An Underactuated Gripper with Grasshopper-Inspired Linkages and Delay Triggering for Pinching and Adaptive Grasping
H. DING†, Y. CHEN†, D. JIA, W. ZHANG* — *IEEE ICIA 2025*

[P7] An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping
Y. Chen, H. Luan, **H. Ding**, W. Zhang* *IEEE RAAI 2025*

Other Publications

[P8] Recent Progress on the Research of 3D Printing in Aqueous Zinc-Ion Batteries
Y. Liu†, **H. Ding**†, H. Chen, H. Gao, J. Yu, F. Mo, N. Wang* *Polymers (MDPI), 2025*

[P9] Rapid Analysis of Heavy-Metal Element Adsorption by SCG Based on LIBS Technology
H. Xie, L. You, X. Fang, L. Li, **H. Ding**, Z. Zhou, G. Zhang, D. Zhang *E3S Web of Conferences, 2025*

Honors & Awards

- 2025 **Undergraduate Champion**, ASME Student Mechanism & Robot Design Competition *USA*
- 2025 **National Scholarship**, Ministry of Education of China *China*
- 2024 **President's Award**, Shenzhen Technology University *Shenzhen, China*
- 2024 **Research and Innovation Award (First Prize)**, Shenzhen Technology University *Shenzhen, China*
- 2023 **Gold Prize**, 9th China "Internet+" Innovation Competition (Guangdong Division) *Guangdong, China*
- 2023 **Second Prize**, 17th "Challenge Cup" Extracurricular Science Competition (Guangdong Division) *Guangdong, China*
- 2023 **Research and Innovation Award (First Prize)**, Shenzhen Technology University *Shenzhen, China*

Professional Service

- 2025 **Conference Reviewer**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- 2025 **Session Chair**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- 2025 **Conference Reviewer**, IEEE International Conference on Automation Science and Engineering (CASE)